Database Design Documentation

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1. Database Schema

1.1. Provinces

Stores master data for the provinces of Indonesia.

Column	Type	Null	Default	Key	Description
id	BIGINT	NO	AUTO	PK	Primary key
code	VARCHAR(10)	NO	-	UNIQUE	Province code (BPS), e.g., 31 = DKI Jakarta
name	VARCHAR(100)	NO	-	-	Province name
is_active	BOOLEAN	NO	TRUE	-	Active flag. Required filter in the response
is_deleted	BOOLEAN	NO	FALSE	-	Soft delete flag
created_at	DATETIME	NO	-	-	Creation time (UTC)
created_by	VARCHAR(50)	YES	-	-	Creator's username
updated_at	DATETIME	YES	-	-	Last update time
updated_by	VARCHAR(50)	YES	-	-	Updater's username
deleted_at	DATETIME	YES	-	-	Soft delete time
deleted_by	VARCHAR(50)	YES	-	-	Deleter's username
version	BIGINT	NO	0	-	Optimistic lock

Index:

- PK_provinces (id)
- UK_provinces_code (code) – UNIQUE
- idx_provinces_name (name) – for filter province name
- idx_provinces_active (is_active, is_deleted) – mandatory filter

1.2. Branches

Stores master data for distribution branches. Each branch is responsible for a few stores.

Column	Type	Null	Default	Key	Description
id	BIGINT	NO	AUTO	PK	Primary key

Column	Type	Null	Default	Key	Description
province_id	BIGINT	NO	-	FK to provinces.id	Province where the branch is located
code	VARCHAR(20)	NO	-	UNIQUE	Unique branch code
name	VARCHAR(150)	NO	-	-	Branch name
address	TEXT	YES	-	-	Full address
phone	VARCHAR(20)	YES	-	-	Phone number
is_active	BOOLEAN	NO	TRUE	-	Active flag
is_deleted	BOOLEAN	NO	FALSE	-	Soft delete flag
created_at	DATETIME	NO	-	-	Creation date and time
created_by	VARCHAR(50)	YES	-	-	Creator's username
updated_at	DATETIME	YES	-	-	Update date and time
updated_by	VARCHAR(50)	YES	-	-	Updater's username
deleted_at	DATETIME	YES	-	-	Soft delete date and time
deleted_by	VARCHAR(50)	YES	-	-	Username of the user who deleted it
version	BIGINT	NO	0	-	Optimistic lock

Index:

- PK_branches (id)
- UK_branches_code (code) – UNIQUE
- FK_branches_province (province_id)
- idx_branches_name (name)
- idx_branches_active (is_active, is_deleted)

Foreign Key Behavior: ON DELETE RESTRICT - A province cannot be deleted if it still has branches.

1.3. Stores

Stores the store's master data. The current volume is approximately 20,000 and continues to grow.

Column	Type	Null	Default	Key	Description
id	BIGINT	NO	AUTO	PK	Primary key
branch_id	BIGINT	NO	-	FK to branches.id	Responsible branch
code	VARCHAR(20)	NO	-	UNIQUE	Unique store code
name	VARCHAR(150)	NO	-	-	Store name
address	TEXT	YES	-	-	Full address
is_whitelisted	BOOLEAN	NO	FALSE	-	Whitelist flag

Column	Type	Null	Default	Key	Description
is_active	BOOLEAN	NO	TRUE	-	Active flag
is_deleted	BOOLEAN	NO	FALSE	-	Soft delete flag
created_at	DATETIME	NO	-	-	Creation time
created_by	VARCHAR(50)	YES	-	-	Creator's username
updated_at	DATETIME	YES	-	-	Update time
updated_by	VARCHAR(50)	YES	-	-	Updater's username
deleted_at	DATETIME	YES	-	-	Soft delete time
deleted_by	VARCHAR(50)	YES	-	-	Username of the user who deleted it
version	BIGINT	NO	0	-	Optimistic lock

Index:

- PK_stores (id)
- UK_stores_code (code) – UNIQUE
- FK_stores_branch (branch_id)
- idx_stores_name (name)
- idx_stores_active (is_active, is_deleted)
- idx_stores_whitelist (is_whitelisted, is_active, is_deleted)

Foreign Key Behavior: ON DELETE RESTRICT - A branch cannot be deleted if it still has a store.

1.4. Users

Column	Type	Null	Default	Key	Description
id	BIGINT	NO	AUTO	PK	Primary key
username	VARCHAR(50)	NO	-	UNIQUE	Username login
password	VARCHAR(255)	NO	-	-	BCrypt hash (cost 10)
role	VARCHAR(20)	NO	-	-	ADMIN / MANAGER / VIEWER
is_active	BOOLEAN	NO	TRUE	-	Active flag
is_deleted	BOOLEAN	NO	FALSE	-	Soft delete flag
created_at	DATETIME	NO	-	-	Creation time
created_by	VARCHAR(50)	YES	-	-	Creator's username
updated_at	DATETIME	YES	-	-	Update time
updated_by	VARCHAR(50)	YES	-	-	Updater's username
version	BIGINT	NO	0	-	Optimistic lock

Index:

- PK_users (id)

- UK_users_username (username) – UNIQUE

CHECK Constraint: role IN ('ADMIN', 'MANAGER', 'VIEWER')

1.5. Audit_logs

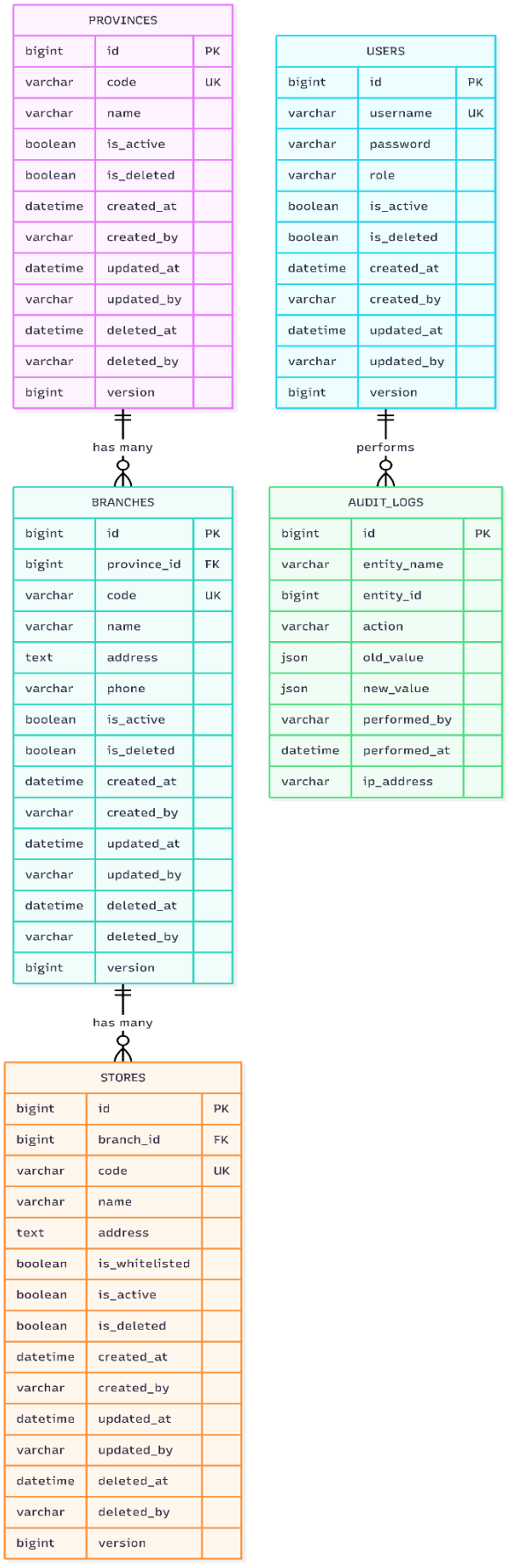
Maintain an audit trail of CUD operations. Append-only (no updates or deletions allowed).

Column	Type	Null	Default	Key	Description
id	BIGINT	NO	AUTO	PK	Primary key
entity_name	VARCHAR(50)	NO	-	-	Entity name, e.g., Store
entity_id	BIGINT	NO	-	-	ID of the modified record
action	VARCHAR(20)	NO	-	-	CREATE / UPDATE / DELETE
old_value	JSON	YES	-	-	Snapshot before the change (NULL for CREATE)
new_value	JSON	YES	-	-	Snapshot after the change (NULL for DELETE)
performed_by	VARCHAR(50)	NO	-	-	Actor username
performed_at	DATETIME	NO	-	-	Operation time
ip_address	VARCHAR(45)	YES	-	-	Actor's IPv4/IPv6 address

Index:

- PK_audit_logs (id)
- idx_audit_entity (entity_name, entity_id) – for per-record traces
- idx_audit_performed (performed_at) – for periodic reports

2. Entity Relationship Diagram



3. Relationship between Tables

3.1. Provinces to Branches (1: N)

A province can have multiple branches. Each branch must be in a single province.

- **FK**: branches.province_id → provinces.id
- **On Delete**: RESTRICT

Deleting a province that still has branches would result in orphaned branch data. The RESTRICT constraint forces administrators to either move or delete the branches first (using a soft delete, so the data remains safe).

3.2. Branches to Stores (1: N)

A single branch manages multiple stores. Each store must belong to a single branch.

- **FK**: stores.branch_id → branches.id
- **On Delete**: RESTRICT

Same as point 3.1. This also maintains the integrity of the hierarchy: every store must be traceable to its branch and province.

3.3. Users to Audit Logs (1: N)

Each user can have multiple audit logs. Each log is recorded under a single user's name.

- **Relation:** Logical (not a physical foreign key). The `audit_logs.performed_by` column stores the username as a snapshot

Audit logs must be immutable and remain valid even if a user is deleted or renamed. A physical foreign key would make the logs vulnerable to changes in user data. This is an intentional trade-off.